Profit (₹ in Crore)	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Number of Companies	8	12	20	30	20	10

Calculate the standard deviation of the profit earned.

Solution:

A = 35 C = 10

Profit (Rs. in Crore)	Mid-value (x)	$d = \frac{x-A}{C}$	f	fd	fd^2
0 – 10	5	-3	8	-24	72
10 – 20	15	-2	12	-24	48
20 – 30	25	-1	20	-20	20
30 – 40	35	0	30	0	0
40 – 50	45	1	20	20	20
50 – 60	55	2	10	20	40
Total			**100**	**-28**	**200**

$$\text{Standard deviation } \sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times C$$

$$= \sqrt{\frac{200}{100} - \left(\frac{-28}{100}\right)^2} \times 10$$

$$= \sqrt{2 - (0.078)} \times 10$$

$$= 13.863$$

Activity

Find the standard deviation for this problem using the other two formulae.

6.4 Combined Mean and Combined Standard Deviation

Combined arithmetic mean can be computed if we know the mean and number of items in each group of the data.

$\overline{x_1}, \overline{x_2}, \sigma_1, \sigma_2$ are mean and standard deviation of two data sets having n_1 and n_2 as number of elements respectively.

$$\text{combined mean } \overline{x_{12}} = \frac{n_1\overline{x_1} + n_2\overline{x_2}}{n_1 + n_2} \quad \text{(if two data sets)}$$

$$\overline{x_{123}} = \frac{n_1\overline{x_1} + n_2\overline{x_2} + n_3\overline{x_3}}{n_1 + n_2 + n_3} \quad \text{(if three data sets)}$$

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